Paired Clock Time Transfer from Starlink over Wi-Fi: Causal Filtering, Drift, and GNSS-Shadow Validation

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Abstract

We test whether ordinary Wi-Fi beacon timing adds useful clock information to Starlink terminal NTP when AP UTC alignment and discipline are unknown. AP-assisted estimates are compared with an AP-free NTP/local baseline and matched controls. GNSS-shadow PPS supplies each method's fixed prior phase base and a validation witness, never a frequency correction. Two SL027 windows each allocate 900 seconds to training and 900 to future evaluation, with filter parameters locked using earlier data. Continuous estimates retain a 60-second source lease; longer frozen forecasts are research holdover. On all 900 future PPS per window, the nominated robust AP model reduces prior-phase-corrected RMS from the AP-free control's 37.180 to 19.226 μs and from 35.662 to 8.240 μs. The first gain reduces bias while slightly worsening centered scatter; the second reduces both. Ordinary AP fits and servos have mixed outcomes. Raw phase RMS remains above 200 μs; earlier failures and incomplete histories remain separate. AP information can differ from the local clock, but constant-rate cancellation does not remove AP wander, NTP bias or asymmetry. These externally calibrated PPS-input comparisons do not establish portable GNSS-free UTC accuracy or measured output.

1 Question, clock domains and prior work

Starlink describes its terminal NTP as GPS-based and best-effort, with no accuracy guarantee [Starlink, n.d.]. The access point broadcasts TSF time, but we have not measured its discipline relative to the terminal. Terminal, AP, client MAC, local timer and capture counter form observable clock domains; their physical dependencies remain unresolved.

Deployment requires an existing dish/AP and a client exposing receive-event timestamps. Our client is an ESP32-S3 W12; ordinary phone/laptop APIs need not expose equivalent information. Qualification adds GNSS hardware. The measured phase base is not portable across clients, paths, boots or models.

The motivating beacon record gave 0.409 μs causal relative RMS over 600 seconds, with one missing raw record and a failed no-gap gate. A separate shadow-PPS observation showed about 6.447 ppm AP gain: small relative residuals do not establish GPS alignment. Espressif specifies microsecond timestamp units and sleep conditions, not RF endpoint bias or UTC accuracy [Espressif Systems, n.d.].

Delay filtering and clock discipline are established [Mills et al., 2010, chrony project, 2025]. Averaging trades network-jitter suppression against oscillator-wander tracking [NTP Project, 2022]. Two-timescale designs still cannot partition clock/network noise without ancillary information [Levine, 2016]; even ideal affine clocks have offset/directional-delay ambiguities [Freris et al., 2011]. This is a paired experiment, not a first-use, optimal-filter or accuracy-ranking claim.

2 Causal observations and paired models

2.1 NTP and the two clock coordinates

The client timestamps immediately before connected UDP `send` and after `recv` returns, giving local T_1, T_4; NTP supplies server receive and transmit T_2, T_3. The conventional estimates are [Mills et al., 2010]

$$\theta_i = \tfrac{1}{2}[(T_2 - T_1) + (T_3 - T_4)], \qquad \delta_i = (T_4 - T_1) - (T_3 - T_2). \tag{1}$$

These software endpoints retain stack, scheduling and directional-transfer bias; subtracting server residence does not identify it. The requests address the local terminal service, not satellite RF timing. Admission checks protocol,

transaction, context and timing bounds. Eight requests about one second apart per 16 seconds yield one lowest-RTT observation per complete burst, newer replies winning ties. Refused/incomplete bursts are retained. Runtime `MIN_RTT8` instead chooses the lowest-delay valid anchor from the latest eight accepted exchanges; neither policy implements full NTP discipline.

Let L denote separately read local-timer microseconds, A AP TSF microseconds, q raw 80 MHz capture ticks and U the NTP UTC coordinate. A prior affine map connects q to L. Separate prior maps connect q to the client MAC receive counter and then to AP beacons. For an NTP endpoint, the AP map must commit before the independent local-map read begins, and that read must precede the endpoint. The certified map is inverted to place the endpoint in q, then projected into A. Exact split midpoint coordinates retain half-microsecond parity; integer UTC/AP origins avoid subtracting large floating-point epochs. All input replies must already have arrived before model publication.

The AP/local pair fits the *same AP-mappable burst minima* with identical weights. For coordinate $X \in \{L, A\}$, the matched affine model is

$$\begin{aligned} \widehat{U}_X(X) &= U_0 + a_X + k_X(X - X_0), \\ (a_X, k_X) &= \arg\min_{a,k} \sum_i w_i [U_i - U_0 - a - k(X_i - X_0)]^2, \qquad w_i = \max(\delta_i, 1\,\mu\text{s})^{-2}. \end{aligned} \tag{2}$$

Here k_X is a dimensionless UTC-microseconds/coordinate-microseconds ratio. Inverse-squared RTT is a declared heuristic, not a measured inverse variance. A separate all-NTP/local control retains minima whose AP endpoint is unavailable. The primary system comparison uses the nominated robust AP model against this AP-free NTP/local 900-second WLS baseline, on identical future witnesses and with each estimator's own prior fixed phase base. Weighting and coordinate can both differ; any gain is not uniquely an AP-oscillator effect. The shared-Huber local model is a secondary mechanism control: its AP-derived weights mean that it is not AP-free. Where AP-mappable input populations differ, their selection effect is also reported separately.

2.2 What the AP can add

If $A = \alpha + \beta L$ exactly over training and prediction, matched affine fits in A and L are algebraically equivalent. Cancelling constant AP drift adds no information; potential value comes from non-affine stability/noise. The exact dual-rate relation is

$$\frac{dU}{dL} = \frac{dU}{dA}\frac{dA}{dL}. \tag{3}$$

Slow NTP/AP calibration estimates dU/dA; frequent beacons/maps track dA/dL. A steadier AP may attenuate local-rate changes, but AP wander leaks until the slow loop responds. Exact AP/local subtraction returns the local coordinate; finite-bandwidth smoothing cannot guarantee zero AP inheritance. Client MAC independence from AP discipline and the RF timestamp endpoint remain uncalibrated.

2.3 Admission, availability and phase scoring

History is bounded to 900 seconds and 128 minima. At least eight points spanning 60 seconds are needed. A prior *unweighted* local OLS model gates a new minimum. For its earlier residuals r_k, set $s = 1.4826\,\text{median}\,|r_k - \text{median}(r)|$; the threshold is

$$G_i = \max(50\,\mu\text{s}, 6s) + \tfrac{1}{2}\max(0, \delta_i - \delta_{\text{min,prior}}). \tag{4}$$

The final fit uses the declared RTT weights. Three consecutive rejected minima clear history and require reacquisition. The gate can reject a true clock step; smooth contamination and early points admitted before initialization can pass. A separate 50 μs beacon-prediction innovation withdraws AP source context. It does not identify the cause or detect every step or smooth wander.

GNSS PPS is captured in q and projected using a snapshot committed strictly before the physical edge, with a 250 ms bridge lease and a fix known before it. The raw witness is retained even if a map is unavailable. AP unavailability means *unknown*, never a substituted local coordinate or zero error. A network-generation change resets models; an AP-population change resets paired AP-dependent history. Capture-map epochs cannot identify AP clock corrections.

For each window, the AP-available members of the last 16 qualified pre-cutoff PPS form one identical calibration identity set, with at least eight required. Each method has its own fixed base B_X. Frozen models retrodict these earlier PPS with their sealed final fit; continuous methods use only their actual prior predictions at those edges. Subsequent error is

$$e_j = \text{phase}_{1\text{s}}[\widehat{U}_X(X_j)] - B_X. \tag{5}$$

No GNSS frequency, future error or validation mean trains a correction. Raw phase is reported separately: an installation without the measured GNSS base does not inherit phase-corrected RMS. Future recentering/detrending would be a different metric and never changes the primary score. All future tails remain. Modulo-one-second phase cannot itself detect an integer UTC-second error. PPS association, receiver/wiring delay, RF endpoint bias and electrical output uncertainty are not independently calibrated.

3 Protocol and locked offline filter experiment

Two SL027 windows, R1–R2, use one image/boot and fixed request cadence. Wi-Fi power saving and automatic light sleep are disabled. At least 300 seconds of full-logging warmup precede the first 1800-second window. Each resets paired fitting, trains a native model for 900 seconds, then retains 900 future PPS. Frozen coefficients, inputs and phase bases are sealed before future inspection. Eight-second expiry audits omit complete beacon/map history; 30 seconds of recorded reprime follow. Missing history remains unknown. A pre-access collector startup failure supplies no timing observations. Frozen source age beyond 60 seconds is research holdover, not continuous service.

Filters/nominations were developed on historical SL025 P1 and locked before P2/P3 heldout errors. Exact estimator code and parameters carry into both fresh windows. After R1, the author reduced the original three-window plan to two before R2 freeze/future scoring; R3 is not acquired, not a failed/missing trial. Plan and amendment remain recorded; no filter is retuned or selected from SL027.

The semantic adapter excludes only isolated same-context refused AP maps from the method view. They supply no point or renewed lease. Explicit reset/new-epoch barriers and every raw/native-invalid record remain audited. An unfenced reset bars a verified adapted comparison; no unavailable map is invented. All declared candidates remain in validation:

- RTT-weighted affine fits over trailing 300 or 900 seconds, in matched local/AP coordinates, plus a separate all-NTP/local control.
- One robust 900-second pair: initial RTT WLS followed by five fixed Huber reweighting iterations, $c = 1.5$, with residual scale $\max(10\,\mu\text{s}, 1.4826\,\text{MAD})$. AP-derived factors $\min(1, 1.5s/|r_i|)$ multiply RTT weights and are shared exactly with the local fit. This is *AP-assisted residual weighting*, not NTP-only filtering. Every retained point has positive weight; no hand-picked point is deleted.
- Two phase-continuous local/AP servo pairs, with 60 or 180-second phase time constants, 300-second frequency averaging and a 10 ppm phase-slew cap.

Targets publish only after completed bursts at a subsequent committed map with the prior AP certificate. Continuous source age uses the latest admitted reply actually retained by that target; the 900-second history never extends the 60-second lease.

Servos integrate split UTC in q with $dU/dq = (dU/dX)(dX/dq)$, preserving phase at updates. They average target rate over 300 seconds and add a 10 ppm-capped phase correction $e/(10^6\tau)$, where e is microseconds and $\tau = 60$ or 180 seconds. Three target disagreements above $500\,\mu$s withdraw/reacquire. AP absence still withholds qualification. Software continuity is not physical output; computation/publication worst-case latency is unmeasured. Huber AP900 and AP servo60 were nominated from P1; the latter was not a development benefit. Synthetic step/spike/continuity tests are distinct from field accuracy evidence.

3.1 Client repairs and availability cohorts

SL027 changes acquisition reliability, not these estimators. Four published bridge slots plus one staging slot place coefficient copies before the availability latch; only successful commits train/increment serials. Refusals retain the prior model under its unchanged 250 ms lease. Each scheduled 100 ms update allows one bounded fresh-read retry, never across a counter/source fault. Integer continuity arithmetic preserves the $1000\,\mu$s fence. A fresh counter sample after each PPS dequeue repairs the newer-queued-edge/older-current race. Reset/refusal telemetry distinguishes withdrawal from retained history.

SL026 qualification retains 600 physical/AP and 588 common projections, with 51 AP refusals and zero replay mismatches/new fault increments. Its 98.0% availability fails the 98.1% gate. Relative to the declared historical 344/1799 baseline, the descriptive unavailable-fraction reduction is 89.54%, not a population failure bound. The author accepted

Table 1: Prior-phase-corrected errors against shadow PPS in μs. NTP-only WLS and nominated robust AP hybrid use identical scored witnesses; ordinary AP WLS and the servo retain their declared paired populations. Each method uses its own fixed phase base on the same prior PPS subset, with no validation recentering or detrending.

Run	Method	n	Mean	Scatter	RMS	Abs. P99	Max. abs.
R1	NTP-only WLS	900	-34.01	15.02	37.18	67.33	67.66
R1	Robust AP	900	11.44	15.45	19.23	43.08	43.85
R1	AP WLS	900	-41.55	17.02	44.90	79.80	80.73
R1	AP servo 60	900	-47.04	16.26	49.77	75.61	75.66
R2	NTP-only WLS	900	-32.61	14.43	35.66	54.77	55.21
R2	Robust AP	900	-4.79	6.70	8.24	20.07	20.24
R2	AP WLS	900	-17.70	12.04	21.40	40.78	40.93
R2	AP servo 60	900	-2.22	8.97	9.24	19.77	20.09

Table 2: Continuously updated offline methods, fixed own prior phase bases. Each cell is matched local/AP RMS in μs on the AP-common population. All primary runs use the locked candidates unchanged; historical exploratory P1 remains in the selected data. Huber weights are AP-derived and shared, so its local branch is also AP-assisted. No validation winner is selected.

Method	R1	R2
WLS 300 s	82.08/82.74	61.16/60.94
WLS 900 s	37.18/44.90	33.91/21.40
Shared-AP Huber 900 s	24.46/19.23	20.40/8.24
Continuous servo 60 s	38.36/49.77	22.13/9.24
Continuous servo 180 s	40.82/56.79	14.91/8.51
All-NTP WLS 300 s	82.08	61.82
All-NTP WLS 900 s	37.18	35.66
Original MIN8	473.63	174.22

Single values use all-NTP inputs, scored on the same AP-common PPS subset.

that availability to proceed; this neither passes the old gate nor qualifies the distinct SL027 image/boot. No additional 600-second qualification precedes the new cohort.

4 Measured comparisons and retained coverage

The tables retain native failures and use the same physical PPS for each local/AP comparison. Signed mean, population-centered standard deviation, RMS, nearest-rank absolute P99 and maximum describe fixed-base PPS agreement. RMS is $\sqrt{\mu^2 + \sigma^2}$; it does not become an uncertainty bound because many correlated PPS were collected. A descriptive future fit $e_j = c + d\tau_j + \eta_j$ reports drift d in microseconds per local second (ppm) and detrended RMS. Neither c nor d corrects the accuracy scores [Riley, 2008]. A replayed window remains valid regardless of RMS; unavailable/unverified states are separate quality classifications. Large future errors are never discarded to satisfy a timing target.

4.1 Two fresh locked-validation windows

Each window retains all 1800 physical PPS, with independently replayed local/AP maps and NTP anchors. All 13 continuous methods have 900/900 verified future predictions and prior phase-base histories, with no gaps, mismatches, new fault increments or bridge resets. R1/R2 retain 97/122 AP and 895/779 local read refusals without losing these projections. One unpaired pre-window NTP record remains unknown but enters no retained dependency. Availability is an observed finite-window result, not a reliability guarantee.

The primary comparison is nominated robust AP900 versus AP-free all-NTP WLS900 (Table 1, Fig. 1). Prior-phase-corrected RMS falls 37.180 to 19.226 μs in R1 and 35.662 to 8.240 μs in R2, on the same 900 future PPS per window. In R1, mean changes from -34.013 to $+11.441$ μs, but centered scatter increases from 15.016 to 15.451 μs: the RMS gain is bias reduction, not jitter suppression. R2 reduces both mean magnitude (-32.615 to -4.790 μs) and scatter

Table 3: Training residual scatter includes NTP source, path, client endpoints and coordinate effects. It cannot isolate source-only jitter. Minima refer to the canonical frozen fit, before Huber downweighting.

Run	Accepted NTP (ms)	Local minima (μs)	AP minima (μs)	Gate rejects
R1	3.897	217.11	217.38	0
R2	4.556	102.74	102.97	0

Table 4: Frozen forecasts after 900 s training: PPS-relative errors after the fixed prior phase base. All 900 future opportunities remain in each run. Beyond true source age 60 s these are research holdover, not qualified UTC service.

Run	Known/900	AP RMS	Local RMS	Fresh n	Fresh AP/local RMS
R1	900/900	35.168	65.843	44	1.977/2.923
R2	900/900	31.777	50.824	58	2.146/3.999

All RMS entries are in μs; fresh means true source age < 60 s.

(14.425 to 6.704 μs). Its mean-square improvement comprises 86.45% reduced squared bias and 13.55% reduced variance. These are two correlated same-boot windows, not a general performance distribution.

The complete matrix retains reversals (Table 2). Ordinary AP900 is worse than AP-free WLS900 in R1 (44.904 versus 37.180 μs), better in R2 (21.404 versus 35.662 μs). The nominated AP servo60 similarly worsens its matched local control in R1 but improves it in R2. Shared-Huber local RMS is 24.462/20.397 μs; those AP-derived weights make it a secondary mechanism control, not an AP-free alternative. R2 matched-local WLS900 differs from all-NTP WLS900 (33.909 versus 35.662 μs), exposing input-selection effects as well. No validation winner replaces either nominee.

Raw phase RMS for AP-free/robust AP is 224.160/217.046 μs in R1 and 226.778/218.521 μs in R2. Each method's own 16-PPS prior base produces the corrected scores; lower corrected RMS does not deliver that accuracy without external phase calibration. There is no future mean subtraction.

Training accepts 448/449 exchanges in R1/R2, yielding 56 complete burst minima per run. Native logs count 0/1 discarded training partials; offline candidate admission counters report 0/0 and remain separate. Both retain 56 training minima. R2 future admission retains 56 all-NTP versus 55 paired minima, with one AP-mapping omission. The canonical prior-fit gate rejects none and causes no reacquisition: real spike-rejection benefit is unproven. Huber downweighting is a separate positive-weight operation. The observed all-exchange versus selected-minimum scatter (Table 3) mixes source, path, client and coordinate effects; selection changes its population and cannot identify isolated source jitter or a fit confidence bound.

Native frozen AP/local forecasts give 35.168/65.843 μs and 31.777/50.824 μs RMS across all 900 future PPS (Table 4). Only 44/58 witnesses have true source age below 60 seconds; their AP RMS is 1.977/2.146 μs. Remaining ages extend to 915/901 seconds and represent holdover stress, not continuous service. Post-hoc AP detrended RMS of 0.936/0.367 μs characterizes residual precision after removing future phase/rate; it is not the achieved forecast error or UTC accuracy.

4.2 Historical comparisons and retained quality failures

Historical SL025 P1/P2 retain 727/728 common future PPS, respectively, with 172 AP unknowns each and one additional P1 local failure. Their native-known coordinates, retained/historical inputs and phase bases are independently checked. Four P2 exchanges with initial history gaps enter no retained target.

P1 AP/local WLS900 gives 19.176/11.234 μs RMS; shared Huber gives 7.939/10.230 μs. In locked P2, AP/local WLS gives 11.529/38.774 μs, but robust AP worsens to 18.889 μs. AP/local servo60 changes from 28.987/8.625 μs in P1 to 10.865/44.146 μs in P2. Favorable frozen holdover cannot replace these mixed continuous-service outcomes.

P3 retains 695 native-computed paired opportunities, 204 AP unknowns and one local failure. Ten native-known AP projections and two retained training endpoints remain unverified: one needs unrecorded inter-window data; the other has a short reset epoch unrecognized by the old replay. Unknown fixed-base dependencies persist after rolling inputs expire. Paired900, Huber and servo families thus have zero fully verified calibrated P3 errors; only 455 common short-history errors are fully replayed. All 900 opportunities/native-computed values remain, without promoting those conditional subsets to replication.

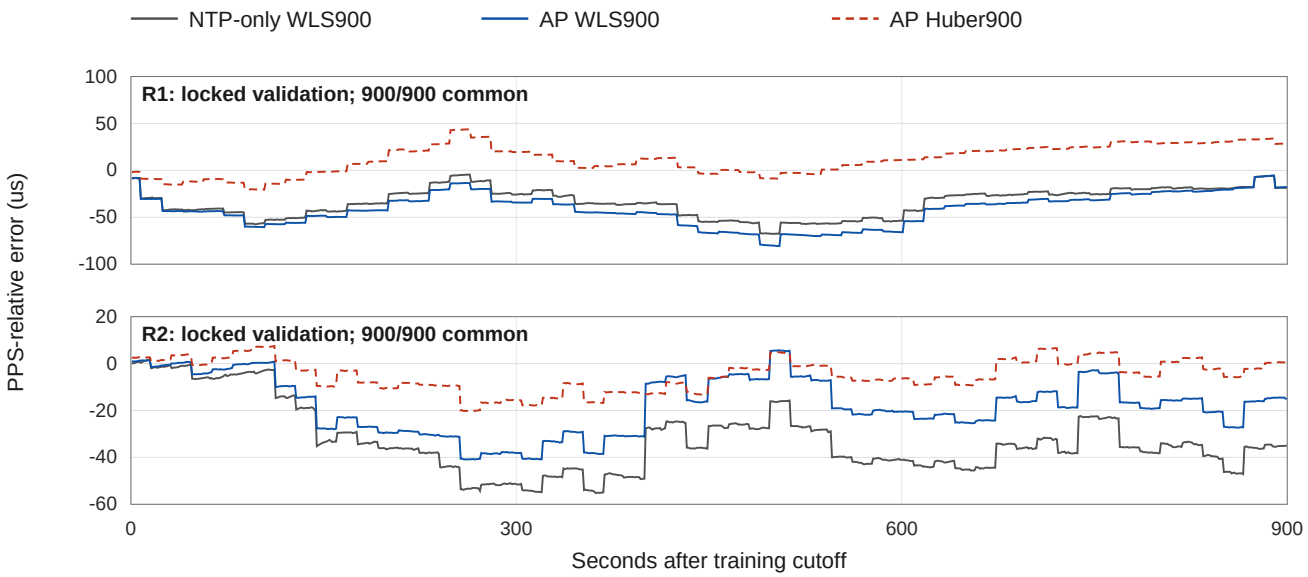


Figure 1: SL027 R1–R2 continuously updated affine phase errors: AP-free all-NTP WLS900, ordinary AP WLS900 and nominated robust AP900, on the same 900 witnesses per window. Each uses its own prior phase base; parameters are locked before fresh data. The robust AP's shared-weight local control is in the matrix. Affine refits can step; phase-continuous servos are not plotted. Vertical scales differ by panel. No future recentering or rate correction is applied.

5 Mechanisms, confounds and earlier failures

Historical P1 GNSS-relative gains change from −0.122 to +0.474 ppm local, but +7.005 to +6.981 ppm AP. Raw-capture versus mapped-local future gain differs by only 0.000112 ppm, limiting a bridge-only explanation. These post-hoc diagnostics never steer a model or locate a physical cause. A frequency-only 1 μs budget over 60 seconds allows just 0.0167 ppm before other uncertainty.

Validation adds three frozen log rows per PPS. Both methods see the same actual load, yet logging, interrupts, temperature, supply and uptime may change rate or software endpoints. Adjacent 60-second parsed-record byte rates increase 9.769% in R1 and 8.072% in R2 at freeze; emission-time binning is approximate, not a CPU/thermal measurement. Zero dropped frames does not establish zero timing impact; no unique logger/source cause is identified.

The old client also creates availability gaps: resetting AP history after a failed read requires 32 accepted maps to reacquire; all 172 P1 AP-invalid PPS lie in that warmup. The P3 queue race retains both raw pulses while refusing conversion as `COUNTER_UNAVAILABLE`. These are client association/history failures, not missing GNSS/USB records or AP correction telemetry. New firmware does not retrospectively repair them or explain smooth drift by itself.

Several phase-error hypotheses remain plausible. Starlink server discipline or software timestamp wander could bias NTP calibration; stratum 1 and zero reported root fields do not measure source uncertainty. Directional queuing and software endpoints can change without a large RTT; minimum delay does not prove reciprocity [Lombardi et al., 2016]. Local oscillator/environment variation can change rate, and small selected samples leave substantial slope leverage [Riley, 2008]. One source/path cannot partition those effects. Frozen models ignore later terminal corrections, whereas continuous targets can inherit them. AP clock behavior is a separate observation, not terminal correction telemetry. Smooth AP wander can pass a phase-innovation threshold.

The earlier direct-NTP trials remain a useful negative control. Three clean SL024 windows retained all 300 future PPS each, with 64.215, 7.795 and 18.737 μs frozen RMS. The first failure is numerically explained by −0.365648 ppm learned-rate mismatch, accumulating about −110 μs in 300 seconds; post-hoc detrending leaves 0.544 μs RMS without repairing the forecast. No deterministic timestamp arithmetic bug sufficient to explain that rate was found. A longer-history model gave 5.424 μs on one repeat's *same* witnesses; a 43-PPS short segment at 0.067 μs was not 67 ns UTC accuracy. The earlier incomplete SL023 near-1 μs record leaves 41 qualified PPS unreplayed and is context only. Original failures, acquisition attempts and operator-interrupted populations remain in the evidence history.

6 Reproduction and qualification limits

The public supplement retains each future opportunity, qualifier, raw prediction and own prior base. A standard-library analyzer recomputes errors and aggregates; it does not recreate acquisition, base-witness measurement or physical bridge/model publication. Private independent replay checks those histories and endpoint certificates, binding selected provenance/coverage. Native frozen models have prospective numeric seals; continuous methods lock their causal calibration rule before validation. Raw logs, identities, credentials and images stay private. Paper/figures/derived data are CC BY 4.0; code/documentation MIT, credited to Jonathan Yantis. Hashes identify bytes, not accuracy.

The GNSS base absorbs setup/model/path phase bias; it is neither portable nor a traceable calibration chain [Lombardi et al., 2019]. Receiver/cable/RMC association, RF-event endpoint, capture/bridge and application mapping remain uncalibrated. No output edge is measured. Clocks may share dependencies, and router/terminal hardware/firmware were not inventoried.

Further qualification needs constant logging load, temperature/load records, reset tests, a calibrated NTP server and independent reference under counterbalanced conditions. Real source steps require an explicit continuity policy. Input availability and physical application/output timing need separate validation.

AP timing is a candidate secondary oscillator, with benefits dependent on calibration, filtering and coverage. These results do not establish repeatable microsecond UTC accuracy or a guaranteed horizon. This remains an unapproved author-review draft, without submission or publication.

References

Starlink. Does Starlink provide time synchronization or NTP servers? Help Center, n.d. Accessed 2 Nov. 2026.

Espressif Systems. ESP-IDF Wi-Fi receive control interface. Pinned v5.5.5 source header, n.d. Commit b774170; accessed 2 Nov. 2026.

D. Mills, J. Martin, J. Burbank, and W. Kasch. Network Time Protocol Version 4: Protocol and Algorithms Specification. RFC 5905, 2010. Accessed 2 Nov. 2026.

chrony project. chrony.conf(5), version 4.8. Configuration manual, 2025. Accessed 2 Nov. 2026.

NTP Project. Clock Discipline Algorithm. Reference implementation documentation, 2022. 23 Nov. 2022 revision; accessed 2 Nov. 2026.

Judah Levine. An Algorithm for Synchronizing a Clock When the Data Are Received Over a Network With an Unstable Delay. *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control*, 63(4):561–570, 2016. doi: 10.1109/TUFFC.2015.2495014. URL `https://tf.nist.gov/general/pdf/2769.pdf`.

Nikolaos M. Freris, Scott R. Graham, and P. R. Kumar. Fundamental Limits on Synchronizing Clocks Over Networks. *IEEE Transactions on Automatic Control*, 56(6):1352–1364, 2011. doi: 10.1109/TAC.2010.2089210. URL `https://cs.nyu.edu/~apanda/classes/sp25/papers/time-sync-limits.pdf`.

W. J. Riley. Handbook of Frequency Stability Analysis. NIST SP 1065, 2008. Accessed 2 Nov. 2026.

M. A. Lombardi, A. N. Novick, G. Neville-Neil, and B. Cooke. Accurate, Traceable, and Verifiable Time Synchronization for World Financial Markets. J. Res. NIST 121; doi:10.6028/jres.121.023, 2016. Accessed 2 Nov. 2026.

M. Lombardi, D. Matsakis, and J. Levine. Metrological and legal traceability of time signals. Inside GNSS; NIST publication record, 2019. Accessed 2 Nov. 2026.